

STAT 4005 - Time Series

Chapter 1 - Introduction

Agenda

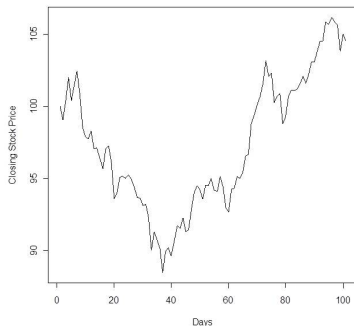
- 1 Basic description
- 2 Decomposition of time series
 - Trends
 - Least squares method
 - Filtering
 - Differencing
 - Seasonal cycles
 - Moving average method
 - Seasonal differencing
- 3 Examples
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Basic description

What is Time Series?

- A collection of values $X_t : t = 1, 2, \dots, n$
 - t : time index (e.g. days, months, years)
 - X_t : observed data at different time (e.g. temperature, stock price)

Example of Time Series :



Distinctive feature of time series

- In classical setting (e.g. t-test, regression), we assume X_t 's are independent
- In time series, we relax the independence assumption
 - Assume X_t 's are dependent
 - The dependence in time series is quantified by **auto-covariance**

For small k , $\text{Cov}(X_t, X_{t+k}) \neq 0$

Some nonstandard features of X_t

1) **Unequally spaced data** ($t = 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 15, \dots$)

- Security Price are not available during non-trading days

2) **Continuous time series** (t can be any real number)

- Physical phenomenon (e.g. temperature)
 - We have $\{X(t), t \in \mathbb{R}\}$ instead of $\{X(t)\}_{t=1,2,\dots}$

3) **Aggregation**

- Data observed : an accumulation of some quantities over a period of time
- Examples:
 - Daily return = aggregation of tick by tick return
 - Daily rainfall = aggregation of hourly rainfall

Some nonstandard features of X_t

4) Replicated time series

- Data observed : repeated measurement of same quantity across different subjects
- e.g.: $X_t^{(j)}$ total weekly spending for the j -th person at week t where $t = 1, 2, \dots, n$ are "repeated measurement" of subject j
- $X_{t_1}^{(j)}$ and $X_{t_2}^{(k)}$ are independent for all t_1, t_2 if $j \neq k$

5) Multiple time series

- $\mathbf{X}_t$ is a vector rather than a scalar
- e.g. Returns of a portfolio at time t : $\mathbf{X}_t = (X_{1,t}, X_{2,t}, \dots, X_{p,t})$
 $X_{i,t}$: return of equity i at time t
- Need to consider the "cross correlation" between different series
e.g. $\text{Cov}(X_{1,t}, X_{2,h})$

Importance of time series

Example 1 - Using correlation to find a "better" confidence interval of μ

Model : $X_t = \mu + a_t - \theta a_{t-1}$ where $a_t \stackrel{\text{iid}}{\sim} N(0, 1)$

- $E(X_t) = \mu + E(a_t) - \theta E(a_{t-1}) = \mu$
- There exists time dependence because

$$\text{Cov}(X_t, X_{t-k}) = \begin{cases} -\theta & k = \pm 1 \\ 1 + \theta^2 & k = 0 \\ 0 & \text{otherwise} \end{cases}$$

Importance of time series

- Approximate 95% confidence interval for the mean μ

$$\bar{X} \pm 2\sqrt{\text{Var}(\bar{X})}$$

- Proof: By Central Limit Theorem (C.L.T.),

$$\frac{\bar{X} - \mu}{\sqrt{\text{Var}(\bar{X})}} \rightarrow N(0, 1)$$

$$\therefore P\left(-2 \leq \frac{\bar{X} - \mu}{\sqrt{\text{Var}(\bar{X})}} \leq 2\right) \approx 0.95$$

$$\Rightarrow P\left(\bar{X} - 2\sqrt{\text{Var}(\bar{X})} \leq \mu \leq \bar{X} + 2\sqrt{\text{Var}(\bar{X})}\right) \approx 0.95$$

Importance of time series

Note that

$$\begin{aligned}\text{Var}(\bar{X}) &= \text{Cov}\left(\frac{1}{n} \sum_{t=1}^n X_t, \frac{1}{n} \sum_{t=1}^n X_t\right) = \frac{1}{n^2} \sum_{t=1}^n \sum_{s=1}^n \text{Cov}(X_t, X_s) \\&= \frac{1}{n^2} (n\text{Cov}(X_1, X_1) + 2(n-1)\text{Cov}(X_2, X_1)) \\&= \frac{1}{n^2} [n(1 + \theta^2) - 2(n-1)\theta] \\&= \frac{1}{n} \left[(1 - \theta)^2 + \frac{2\theta}{n} \right].\end{aligned}$$

Importance of time series

Therefore, a 95% C.I. is

$$\bar{X} \pm \frac{2}{\sqrt{n}} \left[(1 - \theta)^2 + \frac{2\theta}{n} \right]^{1/2}.$$

- Without dependence ($\theta = 0$), 95% C.I. is $\bar{X} \pm \frac{2}{\sqrt{n}}$
- Need to incorporate correlation structure to produce a correct C.I.

Ratio of length of C.I.s is:

$$L(\theta) = \left\{ \frac{2}{\sqrt{n}} \left[(1 - \theta)^2 + \frac{2\theta}{n} \right]^{1/2} \right\} / \left\{ \frac{2}{\sqrt{n}} \right\} = \left[(1 - \theta)^2 + \frac{2\theta}{n} \right]^{1/2}$$

Importance of time series

θ	$L(\theta), n = 50$
-1	1.99
-0.5	1.49
0	1
0.5	0.52
1	0.20

Implication:

- If $\theta = 1$, but we wrongly assumed independence ($\theta = 0$), then our C.I. is "too big"
- If $\theta = -1$, but we wrongly assumed independence ($\theta = 0$), then our C.I. is "too small"
- Need to incorporate correlation structure to produce a correct C.I.

Exercise

Model : $X_t = \mu + Z_t - 0.5Z_{t-1}$ where $Z_t \stackrel{\text{iid}}{\sim} N(0, 1)$. Let $(X_1, X_2, X_3, X_4, X_5) = (4, 5, 6, 4, 1)$.

a Find $\bar{X}$.

$$\bar{X} = \frac{4+5+6+4+1}{5} = 4.$$

b Find the autocovariance function $\gamma(k) = \text{Cov}(X_t, X_{t-k})$.

$$\text{Cov}(X_t, X_{t-k}) = \begin{cases} -0.5 & k = \pm 1 \\ 1.25 & k = 0 \\ 0 & \text{otherwise} \end{cases}$$

c Find $\text{Var}(\bar{X})$.

$$\text{Var}(\bar{X}) = \frac{1}{5} \left[(1 - 0.5)^2 + \frac{2 \cdot 0.5}{5} \right] = 0.09$$

d Find the 95% C.I. for μ .

$$(4 - 2 \cdot \sqrt{0.09}, 4 + 2 \cdot \sqrt{0.09}) = (3.4, 4.6)$$

Exercise

Suppose that $X_t = 10 + Z_t + 0.3Z_{t-1}$, $Z_t \stackrel{\text{iid}}{\sim} N(0, 1)$.

- (a) Find $\text{Var}(X_t)$, $\text{Cov}(X_t, X_{t+1})$, $\text{Corr}(X_t, X_{t-1})$ and $\text{Cov}(X_t, X_{t+k})$ for $|k| \geq 2$.
- (b) Let $\bar{X} = \frac{1}{100} \sum_{t=1}^{100} X_t$. Find $\text{Var}(\bar{X})$.
- (c) Give an example (a value of $\bar{X}$) such that for the hypothesis $H_0 : E(X_t) = 10$ verses $H_1 : E(X_t) \neq 10$, the tests
 - 1) accounting for time dependence, and
 - 2) without accounting for time dependence,give different conclusions.

a)

$$\begin{aligned}\text{Var}(X_t) &= \text{Var}(Z_t) + (0.3)^2 \text{Var}(Z_{t-1}) \\ &= 1.09,\end{aligned}$$

$$\begin{aligned}\text{Cov}(X_t, X_{t+1}) &= \text{Cov}(Z_t + 0.3Z_{t-1}, Z_{t+1} + 0.3Z_t) \\ &= \text{Cov}(Z_t, 0.3Z_t) \\ &= 0.3,\end{aligned}$$

$$\begin{aligned}\text{Corr}(X_t, X_{t-1}) &= \text{Cov}(X_t, X_{t-1}) / (\sqrt{\text{Var}(X_t)} \sqrt{\text{Var}(X_{t-1})}) \\ &= \text{Cov}(Z_t + 0.3Z_{t-1}, Z_{t-1} + 0.3Z_{t-2}) / \text{Var}(X_t) \\ &= \text{Cov}(0.3Z_{t-1}, Z_{t-1}) / \text{Var}(X_t) \\ &= 0.275,\end{aligned}$$

$$\begin{aligned}\text{Cov}(X_t, X_{t+k}) &= \text{Cov}(Z_t + 0.3Z_{t-1}, Z_{t+k} + 0.3Z_{t+k-1}) \\ &= 0.\end{aligned}$$

Solution II

- Observe that $\text{Var}(X_t)$, $\text{Cov}(X_t, X_{t-1})$ are independent of t , and $\text{Cov}(X_t, X_{t-k}) = 0$ for $k \geq 2$. So

$$\begin{aligned}\text{Var}\left(\frac{1}{n} \sum_{t=1}^n X_t\right) &= \frac{1}{n^2} \left(\sum_{t=1}^n \text{Var}(X_t) + \sum_{i \neq j} \text{Cov}(X_i, X_j) \right) \\&= \frac{1}{n^2} \left(\sum_{t=1}^n \text{Var}(X_t) + 2 \sum_{i=1}^{n-1} \text{Cov}(X_t, X_{t-1}) \right) \\&= \frac{1}{n^2} (n \times 1.09 + 2(n-1) \times 0.3) \\&= 0.01684,\end{aligned}$$

where $n = 100$.

Solution III

- Ⓔ) Without assuming dependence, we would have $X_t = 10 + Z_t$ and thus

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{t=1}^n \text{Var}(Z_t) = \frac{1}{n} = 0.01.$$

For example, if $\bar{X} = 9.78$, without accounting for the dependence, the CI is

$$\bar{X} \pm 2\sqrt{\text{Var}(\bar{X})} = 9.78 \pm 2 \times \sqrt{0.01} = (9.58, 9.98),$$

which does not cover 10, thus H_0 is rejected. However, when the dependence is accounted for, the CI is

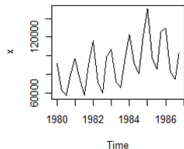
$$\bar{X} \pm 2\sqrt{\text{Var}(\bar{X})} = 9.78 \pm 2 \times \sqrt{0.01684} = (9.52, 10.04),$$

which covers 10, thus H_0 is not rejected.

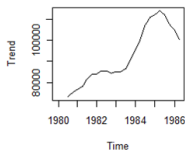
Decomposition of time series

$$X_t = \underbrace{T_t}_{\text{(Trend)}} + \underbrace{S_t}_{\text{(Seasonality)}} + \underbrace{N_t}_{\text{(Noise)}}$$

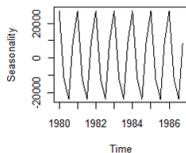
- $T_t + S_t \Rightarrow$ Macroscopic Component
- $N_t \Rightarrow$ Microscopic Component



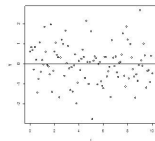
Series



Trend



Seasonality

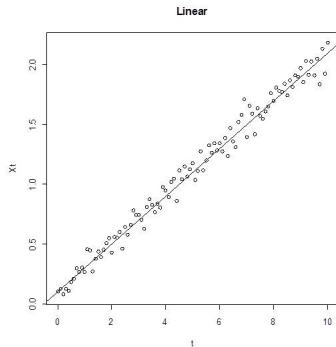


Noise

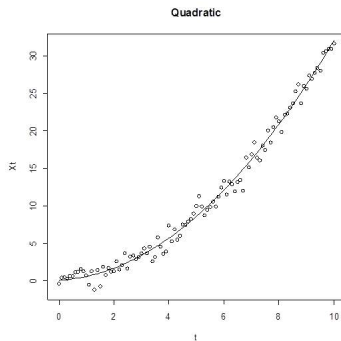
Trends

Examples

- Linear Trend : $T_t = \alpha + \beta t$
- Quadratic Trend : $T_t = \alpha + \beta t + \gamma t^2$



Linear Trend



Quadratic Trend

Estimation of trends without seasonality

When $X_t = T_t + N_t$ (no seasonal effect), the trend can be estimated by

1) Least Squares Method

- minimizes $\sum (X_t - T_t)^2$
- T_t is a simple model of trend, e.g. $T_t = \alpha + \beta t$

2) Filtering

- $\hat{T}_t = S_m(X_t)$
- S_m is a smoother: an operator that computes weighted average of observations near X_t

3) Differencing

- $\Delta X_t = X_t - X_{t-1}$
- Removing trend instead of estimating trend

1. Least squares method

Idea : Find α , β_j s in $T_t = \alpha + \beta_1 t + \cdots + \beta_k t^k$ such that

$$RSS = \sum_{t=1}^n (X_t - T_t)^2 \text{ is minimized}$$

Method: Regression $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e} \Rightarrow \hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$

Comparing $X_t = \alpha + \beta_1 t + \cdots + \beta_k t^k + N_t$ to $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$

$$\Rightarrow \mathbf{Y} = (X_1, \dots, X_n)'; \mathbf{X} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 2 & \cdots & 2^k \\ \vdots & & \vdots & \\ 1 & n & \cdots & n^k \end{pmatrix}$$

$$\Rightarrow \hat{T} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

1. Least squares method

R-Implementation:

```
n=100
k=2
Y=3+2*(1:n)+10*rnorm(n)
X=rep(1,n)
for (j in 1:k) {
  X=cbind(X,(1:n)^(j))
}
Trend.coef=solve(t(X)%*%X,t(X)%*%Y)
Trend=X%*%Trend.coef
ts.plot(Y)
points(1:n,Trend,col=2,type='l')
```

Drawbacks

- Only allow simple form of T_t , otherwise the minimization is difficult

1. Least squares method:

Exercise:

- Given the data $\{1.2, 2.1, 2.9, 3.8\}$, estimate the trend in the form $T_t = \alpha + \beta t$.

$$\mathbf{Y} = \begin{pmatrix} 1.2 \\ 2.1 \\ 2.9 \\ 3.8 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix},$$

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}, \quad (\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{pmatrix},$$

$$\begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \begin{pmatrix} 0.35 \\ 0.86 \end{pmatrix}.$$

2. Filtering

- Idea : Smooth the series using **local data** to estimate the trend

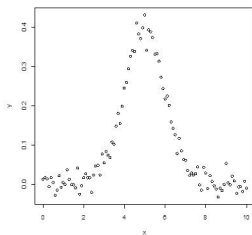
$$\underbrace{\hat{T}_t = S_m(X_t)}_{\text{"smoothed" series}} = \underbrace{\sum_{r=-q}^s a_r X_{t+r}}_{\text{Weighted average of } \{X_{t-q}, X_{t-q+1}, \dots, X_{t+s}\}}$$

- Local: q and s are much smaller comparing to the sample size n .
- The weight $\{a_r\}$ are usually assumed to be symmetric ($a_r = a_{-r}$) and normalized ($\sum a_r = 1$)

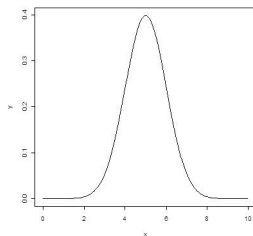
2. Filtering

- Low-pass filter: Filter out high-frequencies (volatile) signals
Retains low-frequencies (smooth) signals

Low-Pass e.g. $S_m(X_t) = \frac{\sum_{r=-q}^q X_{t+r}}{2q+1}$



before Low-pass

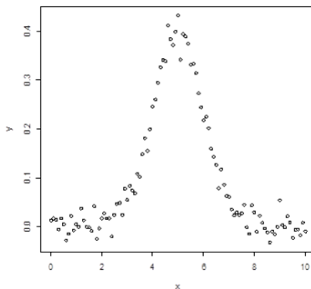


after Low-pass

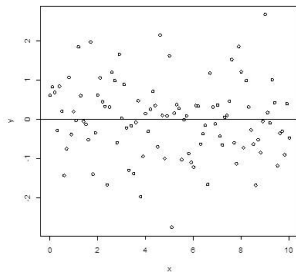
2. Filtering

- High-pass filter: Filter out low-frequencies (smooth) signals
Retains high-frequencies (volatile) signals

High-Pass e.g. $X_t - S_m(X_t)$



before High-pass



after High-pass

2. Filtering

Examples

1) Moving Average Filter

$$\hat{T}_t = \frac{1}{2q+1} \sum_{r=-q}^q X_{t+r}$$

- What is $\{\hat{T}_t\}$ if $\{X_t\} = \{1.1, 2.2, 2.7, 4.1, 5.2, 5.8\}$ and $q = 1$?
- If $X_t = \alpha + \beta t + N_t$ (Trend+Noise),

$$\begin{aligned} \hat{T}_t &= S_m(X_t) = \frac{1}{2q+1} \sum_{r=-q}^q [\alpha + \beta(t+r) + N_{t+r}] \\ &\approx \alpha + \beta t \quad \left(\text{smoothing: } \frac{1}{2q+1} \sum_{r=-q}^q N_{t+r} \approx 0 \right) \end{aligned}$$

2. Filtering

Examples

2) Spencer 15-point filter:

$$(a_0, a_1, \dots, a_7) = \frac{1}{320} (74, 67, 46, 21, 3, -5, -6, -3), \quad a_r = a_{-r}$$

- Property: Does not distort a cubic trend:
 - If $X_t = T_t + N_t$, where $T_t = at^3 + bt^2 + ct + d$, then

$$\begin{aligned}\hat{T}_t = S_m(X_t) &= \sum_{r=-7}^7 a_r T_{t+r} + \sum_{r=-7}^7 a_r N_{t+r} \\ &= at^3 + bt^2 + ct + d + \underbrace{\sum_{r=-7}^7 a_r N_{t+r}} \\ &\approx T_t \quad \left(\sum_{r=-7}^7 a_r N_{t+r} \approx 0 \right)\end{aligned}$$

- We say: the cubic trend "passes through" the filter

2. Filtering

Exercise: Spencer 15-point filter:

$$(a_0, a_1, \dots, a_7) = \frac{1}{320} (74, 67, 46, 21, 3, -5, -6, -3), \quad a_r = a_{-r}$$

What is the filtered/smoothed series if the input is

- $X_t = ct$

$$\begin{aligned} \frac{1}{320} [& (-3) \cdot (-7) + (-6) \cdot (-6) + (-5) \cdot (-5) + 3 \cdot (-4) \\ & + 21 \cdot (-3) + 46 \cdot (-2) + 67 \cdot (-1) + 67 \cdot 1 + 46 \cdot 2 \\ & + 21 \cdot 3 + 3 \cdot 4 + (-5) \cdot 5 + (-6) \cdot 6 + (-3) \cdot 7] = 0. \end{aligned}$$

2. Filtering

- $X_t = bt^2$?

$$\begin{aligned} \frac{1}{320} [& (-3) \cdot (-7)^2 + (-6) \cdot (-6)^2 + (-5) \cdot (-5)^2 + 3 \cdot (-4)^2 \\ & + 21 \cdot (-3)^2 + 46 \cdot (-2)^2 + 67 \cdot (-1)^2 + 67 \cdot 1^2 + 46 \cdot 2^2 \\ & + 21 \cdot 3^2 + 3 \cdot 4^2 + (-5) \cdot 5^2 + (-6) \cdot 6^2 + (-3) \cdot 7^2] = 0. \end{aligned}$$

- $X_t = at^3$?

$$\begin{aligned} \frac{1}{320} [& (-3) \cdot (-7)^3 + (-6) \cdot (-6)^3 + (-5) \cdot (-5)^3 + 3 \cdot (-4)^3 \\ & + 21 \cdot (-3)^3 + 46 \cdot (-2)^3 + 67 \cdot (-1)^3 + 67 \cdot 1^3 + 46 \cdot 2^3 \\ & + 21 \cdot 3^3 + 3 \cdot 4^3 + (-5) \cdot 5^3 + (-6) \cdot 6^3 + (-3) \cdot 7^3] = 0 \end{aligned}$$

2. Filtering

Theorem 1

A k^{th} order polynomial passes through a filter T_t , i.e.

$$\hat{T}_t = \sum_{r=-s}^s a_r T_{t+r} = T_t,$$

for $T_t = c_0 + c_1 t + \dots + c_k t^k$ if and only if

- ① $\sum_{r=-s}^s a_r = 1$
- ② $\sum_{r=-s}^s r^j a_r = 0$ for $j = 1, 2, \dots, k$

2. Filtering

Application: Design a filter such that a quadratic trend would not be distorted.

We design a filter which has order $s = 2$ does not distort a quadratic trend as follows,

$$a_r = a_{-r}$$

$$a_r = (a_0 = 1/4, a_1 = 1/2, a_2 = -1/8)$$

satisfy conditions

$$\sum_r a_r = 1, \quad \sum_r r a_r = 0, \quad \sum_r r^2 a_r = 0.$$

any other filter with different weights $\{a_r\}$ are acceptable.

Exercise

Let $\{a_{-2}, a_{-1}, a_0, a_1, a_2\}$ be a filter and $\{x_1, x_2, \dots, x_{10}\}$ be the observations.

- a) If the filter is applied to the observations, what is the length of the filtered sequence?
- b) Write, in terms of a_k and x_k s, the filtered value for the 6-th observation (x_6).
- c) Suppose that $a_0 = 0.5$, $a_1 = 0.2$, $a_2 = 0.05$ and the filter is symmetric. Does the filter pass through a linear trend without distortion? Does the filter passes through a quadratic trend without distortion?
- d) If $a_0 = 0.5$, $a_1 = a_{-1} = a_2 = a_{-2} = 0.1$, do you think this filter is useful? Explain.

Solution

a) The length of sequence is 6.

b)

$$\hat{x}_6 = a_{-2}x_4 + a_{-1}x_5 + a_0x_6 + a_1x_7 + a_2x_8.$$

c) The filter can pass any linear trend since

$$\sum_{r=-2}^2 a_r = 1,$$

$$\sum_{r=-2}^2 r a_r = 0.$$

But since

$$\sum_{r=-2}^2 r^2 a_r \neq 0,$$

the filter does not pass through quadratic trend.

d) The filter is not useful because $\sum_{r=-2}^2 a_r = 0.9 \neq 1$.

3. Differencing

- Differencing

$$\text{First Order:} \quad \Delta X_t = X_t - X_{t-1}$$

$$\text{Second Order:} \quad \Delta^2 X_t = \Delta(\Delta X_t)$$

- Definition: Backshift operator (B):

- $BX_t = X_{t-1}$

- $B^k X_t = X_{t-k}, k = 1, 2, \dots$

- Definition: Differencing operator (Δ):

- $\Delta X_t = (1 - B)X_t$

- $\Delta^k X_t = (1 - B)^k X_t, k = 1, 2, \dots$

- Exercise: What is $\{\Delta X_t\}$ if $\{X_t\} = \{1.1, 2.2, 2.7, 4.1, 5.2, 5.8\}$

$$\{\Delta X_t\} = \{1.1, 0.5, 1.4, 1.1, 0.6\}.$$

3. Differencing removes trend

- If $X_t = \alpha + \beta t$,
 - $\Delta X_t = X_t - X_{t-1} = \alpha + \beta t - [\alpha + \beta(t-1)] = \beta$ (no Trend!)
- If $X_t = \gamma t^p$,
 - $\Delta X_t = X_t - X_{t-1} = \gamma t^p - \gamma(t-1)^p = p\gamma t^{p-1} - C_2^p \gamma t^{p-2} + \dots$
- In general, if $X_t = T_t + N_t$ and $T_t = \sum_{j=0}^p a_j t^j$,
 - $\Delta^p X_t = p! a_p + \Delta^p N_t$
 - If the trend is a j^{th} degree polynomial, then the trend can be eliminated in differencing j times.

Example

```
n=100
t=1:n
x=5-2*t+3*t^2-4*t^3+10*rnorm(n)
d1=diff(x)
d2=diff(d1)
d3=diff(d2)
d4=diff(d3)
d5=diff(d4)
par(mfrow=c(2,3))
ts.plot(x)
ts.plot(d1)
ts.plot(d2)
ts.plot(d3)
ts.plot(d4)
ts.plot(d5)
```

Seasonal cycles

- General Decomposition

$$X_t = T_t + S_t + N_t$$

- Multiplicative Seasonal Component

$$X_t = T_t S_t N_t$$

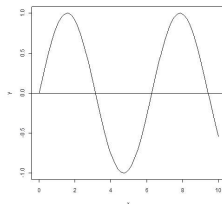
- Seasonal component S_t : period= d

- Requirements

- ① $S_{t+d} = S_t$ (period d : repeating itself after time d)

- ② $\sum_{j=1}^d S_j = 0$ (common effect is explained by T_t)

- e.g. season: $d=4$, month: $d=12$, week: $d=7$



Estimating/Removing seasonal effect

Difficulties when both T_t and S_t exist:

- Need to separate the effect of trend T_t and seasonal effect S_t
- Simple model $T_t = \sum_{j=0}^k a_j t^j$ does not work.
 - ⇒ Cannot use least squares method

Available methods:

- 1) Moving average method
- 2) Seasonal differencing

1. Moving average method

- STEP 1: Estimate the trend T_t by a special moving average filter
 - the moving average filter must be of length d
 - ⇒ the estimated trend is free from seasonal effect because $\sum_{j=1}^d S_j = 0$

$$\hat{T}_t = \begin{cases} \frac{1}{d} \left(\frac{1}{2}X_{t-q} + X_{t-q+1} + \dots + \frac{1}{2}X_{t+q} \right) & , \quad d = 2q \\ \frac{1}{d} \sum_{r=-q}^q X_{t+r} & , \quad d = 2q + 1 \end{cases}$$

1. Moving average method

- STEP 2: Given sample from $t = 1, \dots, nd$, where n is the number of complete cycle and d is the number of seasonal components, we estimate the seasonal components $\hat{S}_j$, $j = 1, \dots, d$, as follows.

$$D_t = X_t - \hat{T}_t,$$

$$\overline{D} = \frac{1}{nd} \sum_{t=1}^{nd} D_t,$$

$$\hat{S}_j = \frac{1}{n} \sum_{k=0}^{n-1} (D_{kd+j} - \overline{D}).$$

- STEP 3: Use any filter for the series $X_t - \hat{S}_t$ to get an improved $\tilde{T}_t$ (can set $\tilde{T}_t = \hat{T}_t$ if you are satisfied with the filter in Step 1.)
- RESULT: $X_t = \tilde{T}_t + \hat{S}_t + \hat{N}_t$, ($\hat{N}_t = X_t - \tilde{T}_t - \hat{S}_t$)

2. Seasonal differencing

- Seasonal differencing

$$\begin{aligned}\Delta^{(d)} X_t &= (1 - B^d) X_t \\ &= X_t - X_{t-d}\end{aligned}$$

- Seasonal differencing removes seasonal effects:

If $X_t = S_t + N_t$ and period $= d$,

then $\Delta^{(d)} X_t = S_t - S_{t-d} + N_t - N_{t-d} = N_t - N_{t-d}$.

(Recall $S_t = S_{t-d}$)

- Drawback - Lose d data points

- data $= (X_1, X_2, \dots, X_n)$
- differenced data $= (\Delta^{(d)} X_{d+1}, \dots, \Delta^{(d)} X_n)$

- No estimated seasonal effect $\hat{S}_t$ is obtained.

Example in R

- Data :
Quarterly operating revenues of Washington power company 80-86
<http://www.sta.cuhk.edu.hk/NHCHAN/TSBook2nd/dataset.html>
- Step 0: Time Series Plot

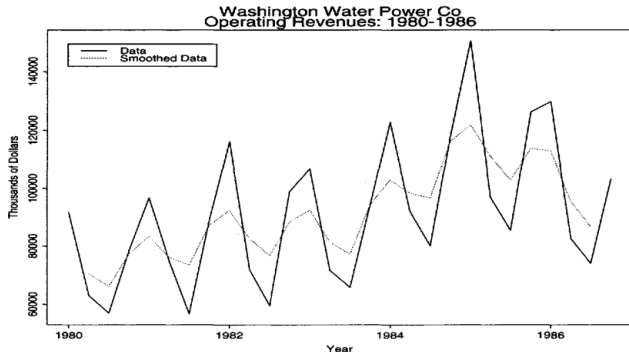


Fig. 1.2 Time series plots.

Step 0: Time series plot

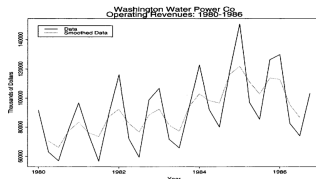


Fig. 1.2 Time series plots.

- Observations

- 1) Slightly increasing trend
- 2) Annual Cycle

- Lower in summer
- Higher in winter (heating?)

R-Codes for reading data and time series plot

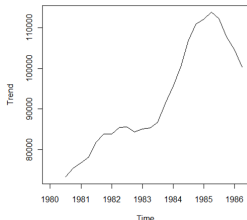
```
x=read.delim("washpower.dat",head=FALSE)
x=ts(x, frequency = 4, start = c(1980, 1))
ts.plot(x,main="Wasington Water Power Co")
```

Decomposition $X_t = T_t + S_t + N_t$: Step 1

- Step 1:

Estimate the trend $\hat{T}_t$ by a filter running over one complete season ($d = 4$)

$$\begin{aligned}\hat{T}_3 &= \frac{\frac{1}{2}X_1 + X_2 + X_3 + X_4 + \frac{1}{2}X_5}{4} \\ \hat{T}_4 &= \frac{\frac{1}{2}X_2 + X_3 + X_4 + X_5 + \frac{1}{2}X_6}{4} \\ &\dots \\ \hat{T}_{26} &= \frac{\frac{1}{2}X_{24} + X_{25} + X_{26} + X_{27} + \frac{1}{2}X_{28}}{4}\end{aligned}$$



```
n=length(x)
# make empty vector to store trend
T1=rep(NA,n)
filter=c(0.125,0.25,0.25,0.25,0.125)
# compute filtered values recursively
radius=2;start=3;end=n-2
for ( k in start:end) {
  T1[k]=filter%*%x[(k-radius):(k+radius)]
}
ts.plot(T1,ylab="Trend")
```

Decomposition $X_t = T_t + S_t + N_t$: Step 2

- Step 2: Estimate seasonal effect from the trend-removed series D_t

$$D_t = X_t - \hat{T}_t, \quad \bar{D} = \frac{1}{24} \sum_{i=3}^{26} D_i$$

Estimating the seasonal part S_i

$$\hat{S}_1 = [(D_5 - \bar{D}) + (D_9 - \bar{D}) + \dots + (D_{25} - \bar{D})]/6$$

$$\hat{S}_2 =$$

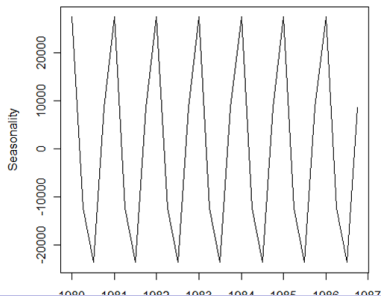
$$[(D_6 - \bar{D}) + (D_{10} - \bar{D}) + \dots + (D_{26} - \bar{D})]/6$$

$$\hat{S}_3 = [(D_3 - \bar{D}) + (D_7 - \bar{D}) + \dots + (D_{23} - \bar{D})]/6$$

$$\hat{S}_4 = [(D_4 - \bar{D}) + (D_8 - \bar{D}) + \dots + (D_{24} - \bar{D})]/6$$

$$\Rightarrow S_{i+4j} = S_i, \quad \text{all } i, j$$

```
d=4;           # set period=4
D=x-T1;
# na.rm=T means remove NA entries
D.bar=mean(D,na.rm=T)
# data of same season put in same col.
S.mat=matrix(D-D.bar,ncol=d,byrow=T)
# compute column(2) mean
S=apply(S.mat,2,mean,na.rm=T)
# make S the same length as x
S=rep(S,n/d)
ts.plot(S,ylab="Seasonality")
```



Decomposition $X_t = T_t + S_t + N_t$: Step 3

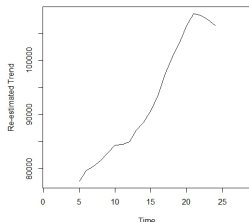
- Step 3: Re-estimate the trend from deseasonalized data $Q_t = X_t - S_t$
Re-estimating the trend by **any** reasonable filter, e.g.

$$\tilde{T}_5 = \frac{1}{9}(Q_1 + \cdots + Q_9)$$

$$\tilde{T}_6 = \frac{1}{9}(Q_2 + \cdots + Q_{10})$$

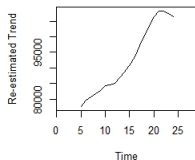
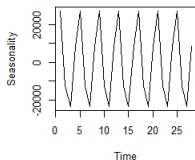
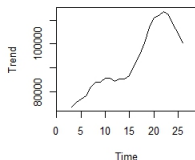
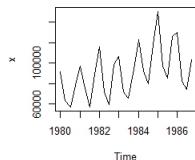
$$\cdots = \cdots$$

$$\tilde{T}_{24} = \frac{1}{9}(Q_{20} + \cdots + Q_{28})$$



```
Q=x-S
# make empty vector to store new trend
newT=rep(NA,n)
filter=rep(1/9,9)
# compute new filtered values recursively
radius=4;start=5;end=n-4
for ( k in start:end) {
  newT[k]=filter*%Q[(k-radius):(k+radius)]
}
ts.plot(newT,ylab="Re-estimated Trend")
```

Putting four graphs together



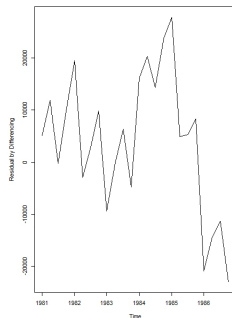
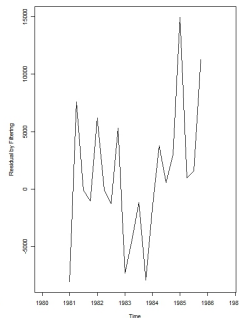
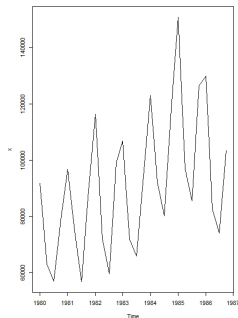
```
par(mfrow=c(2,2)) # make 2x2 plots
ts.plot(x)
ts.plot(T1,ylab="Trend")
ts.plot(S,ylab="Seasonality")
ts.plot(newT,ylab="Re-estimated Trend")
```

- Note that the re-estimated trend is smoother than the trend in Step 1.
- (Again) No need to do Step 3 if satisfied with the trend in Step 1.

Example: Seasonal differencing

$$Y_t = (1 - B^4)X_t = X_t - X_{t-4}$$

```
y=diff(x,4)
par(mfrow=c(1,3))
ts.plot(x)
ts.plot(x-newT-S,ylab="Residual by Filtering")
ts.plot(y,ylab="Residual by Differencing")
```



Decomposition $X_t = \widetilde{T}_t + \widehat{S}_t + \widehat{N}_t$

What to do after decomposition or differencing?

- Check the residual

- $\widehat{N}_t = X_t - \widetilde{T}_t - \widehat{S}_t$ or
- $\widehat{N}_t = X_t - X_{t-d}$

to detect further structure

- Sophisticated models for the microscopic structure in $\widehat{N}_t$ will be discussed in later chapters

Exercise

Use the following R-code to generate a series with trend, quarterly seasonal component and noise:

```
set.seed(4005)
season.effect=c(-20,-10,60,-30)
sea.com=rep(season.effect,20)
t=1:80
trend=-2+0.3*t+0.025*(t^2)
noise=rnorm(80,0,10)
data=trend+sea.com+noise
ts.plot(data)
```

Decompose the series data into three components. Compare the true and the estimated components.

Summary of Chapter 1

1) Time Series

- Observations with time dependent structure
- $\text{Cov}(X_t, X_{t+k}) \neq 0$

2) Decomposition

$$X_t = \underbrace{T_t}_{\text{(Trend)}} + \underbrace{S_t}_{\text{(Seasonality)}} + \underbrace{N_t}_{\text{(Noise)}}$$

i) Removing Trend only

- Least Squares Method: Minimizes $\sum_{t=1}^n (X_t - T_t)^2$
- Filters: $\hat{T}_t = S_m(X_t) = \sum_{r=-q}^s a_r X_{t+r}$
- Differencing $\Delta X_t = (1 - B)X_t = X_t - X_{t-1}$

ii) Removing Trend and Seasonality

- Seasonal Filtering:
 - Step 1: Moving average filter with length d to obtain $\hat{T}_t$
 - Step 2: Estimate $\hat{S}_t$ from $X_t - \hat{T}_t$
 - Step 3: Re-estimate $\tilde{T}_t$ from $X_t - \hat{S}_t$
- Seasonal Differencing $\Delta^{(d)} X_t = (1 - B^d)X_t = X_t - X_{t-d}$